

Survival analysis for infectious disease modelers

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Lorentz Center workshop
Connecting Survival Analysis and Infectious Disease
Modeling
June 15–19, 2026, Leiden



Objective

- ▶ To give a very brief overview of survival analysis
- ▶ For an audience that is familiar with Kaplan-Meier, log-rank test and Cox model
- ▶ Focus on the **principles** and **concepts**
- ▶ How do “we” think?

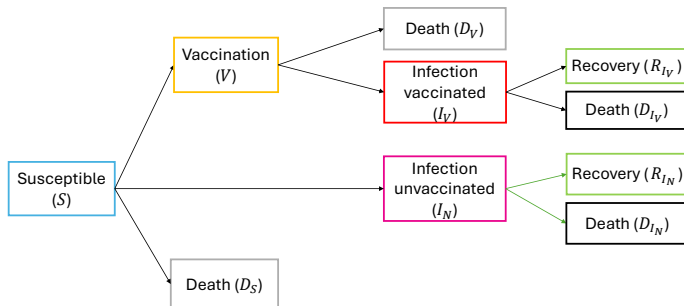
Multi-state models

- ▶ Going to use multi-state models to discuss most of the concepts in survival analysis
- ▶ Why multi-state models?

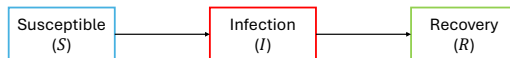
Multi-state models

- ▶ Going to use multi-state models to discuss most of the concepts in survival analysis
- ▶ Why multi-state models?
- ▶ I am a big fan
- ▶ Sufficiently rich in scope to capture most real-life processes
- ▶ Therefore a useful framework (I hope) to connect survival analysis and infectious disease modeling
- ▶ Here are a couple of examples of multi-state models

Example multi-state model



Example multi-state model



Some terminology

- ▶ In the previous picture, a box is called a **state**
- ▶ A direct arrow from one state to another is a **transition**
- ▶ A transition is triggered by an **event** happening
- ▶ State space $\mathcal{S} = \{1, \dots, S\}$, finite
- ▶ Multi-state model is a random process $\{X(t); t \geq 0\}$ taking values in $\mathcal{S}$
- ▶ **History** of the process until s : $\mathcal{H}_s = \{X(u); 0 \leq u \leq s\}$

Standard survival analysis

- ▶ Here we have two states, 0=alive and 1=dead
- ▶ One transition from alive to dead, triggered by the event death
- ▶ Subjects in state 0 are alive, and thereby **at risk** of experiencing the event death
- ▶ For subjects in state 0 there is a probabilistic **intensity**, say $\lambda(t)$, governing event occurrence
- ▶ This intensity is often the primary target for statistical modeling
- ▶ The event of interest will typically not be observed for everyone, due to being event-free at end of follow-up (**censoring**)
- ▶ Or, if the event of interest is not death, by experiencing a **competing event** (like death)

Typical data for standard survival analysis

Typically, the following data can be available for subject i in a sample of n independent individuals:

- ▶ Follow-up time T_i
- ▶ Indicator variable δ_i , indicating whether the event has happened ($\delta_i = 1$) or not ($\delta_i = 0$)
- ▶ A time $V_i < T_i$ where the subject entered the study (if $V_i > 0$ this is called **delayed entry**)
- ▶ A vector of covariate values $Z_i(t) = (Z_{i1}(t), \dots, Z_{ip}(t))$, for $t \in [V_i, T_i]$, some of which may be time-dependent, others time-fixed

Typical data for standard survival analysis

- ▶ From this, one may define a **counting process** $N_i(t) = I\{V_i < T_i \leq t, \delta_i = 1\}$, counting the number of observed events by time t (0 or 1)
- ▶ Defining $dN_i(t) = N_i(t) - N_i(t-)$, observation of the event at time t can then be represented by $dN_i(t) = 1$
- ▶ We can also define the **at risk** process $Y_i(t)$, indicating whether subject i is at risk at time t , i.e.,
 $Y_i(t) = I\{V_i < t \leq T_i\}$

More generally

- ▶ The vector $(T_i, V_i, \delta_i, Z_i(t))$ is “all we need” for standard survival data
- ▶ Multi-state models are more complex
- ▶ But for each of the transitions in the multi-state model, we can still define / construct
 - ▶ Counting process $N_i(t)$, counting the number of observed transitions of subject i by time t
 - ▶ At risk process $Y_i(t)$, indicating whether subject i is at risk for that transition at time t
- ▶ These counting processes and at risk processes should actually be indexed by the transition (not done here)
- ▶ Additional advantage of $N_i(t)$: can be larger than 1 (recurrent events)

Preliminary concepts and issues

Censoring

- ▶ Goal is to estimate quantities in completely observed population
- ▶ “It takes time to observe time” => right censoring
- ▶ We regard patients at risk at t , so with $Y_i(t) = 1$, as representing all patients that would be at risk with complete observation => **independent censoring**
- ▶ Often subject's state is assessed periodically, leading to **interval censoring**
- ▶ Less knowledge about the counting process and about at risk status
- ▶ Here the assumption is that assessment times are independent of the state occupied

Time origin

- ▶ Intensity $\lambda_i(t)$ depends on t (unless it is constant)
- ▶ Time t is measured from a meaningful starting point ($t = 0$)
- ▶ Take $S \rightarrow I \rightarrow R$ model
- ▶ For $S \rightarrow I$, $t = 0$ usually refers to the start of the epidemic (calendar time)
- ▶ For $I \rightarrow R$, in the standard SIR model, $t = 0$ also refers to the start of epidemic (clock forward)
 - ▶ Questionable, since probably more reasonable to take time of infection as $t = 0$ (clock reset)
 - ▶ When the $I \rightarrow R$ intensity is constant, not an issue
- ▶ Sometimes intensity may depend on multiple time scales
 - ▶ Standard to take one as principal, incorporate others as covariates

Delayed entry

- ▶ Also called left truncation
- ▶ May be induced by observation scheme
- ▶ Or may be internal (I $\rightarrow$ R transition with clock forward)
- ▶ Assumption is that time of delayed entry is independent

The intensity

- ▶ Once it has been established who is at risk, what is the event and how t is defined (what is the time origin), we can define the **intensity** for subject i as

$$\lambda_i(t) = \lim_{\Delta t \rightarrow 0} P(dN_i(t) = 1 \mid \mathcal{H}_i(t-)) / \Delta t$$

- ▶ The intensity is intensely useful in survival analysis, because

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$$\lambda_i(t) = \lim_{\Delta t \rightarrow 0} P(dN_i(t) = 1 \mid \mathcal{H}_i(t-)) / \Delta t$$

- ▶ The intensity is intensely useful in survival analysis, because
 - ▶ It gives a dynamical description of how events occur over time, incorporating the past
 - ▶ Estimating the intensity is possible since it relies exclusively on the information of subjects still at risk
 - ▶ We can examine the effect of both **baseline** and **time-dependent** covariates
 - ▶ Delayed entry is naturally taken into account

Hazard models

- ▶ In order to make a **model** for the intensity we need to specify
 - ▶ How it depends on time t
 - ▶ And on the available information in $\mathcal{H}_i(t)$, in particular how it depends on the covariates
- ▶ Let's first avoid making any models and focus on **non-parametric** estimation of the hazard

Estimation of intensities

Non-parametric estimate of the cumulative intensity

- ▶ Increment at event time t_j :

$$\Delta\hat{\Lambda}(t_j) = dN(t_j)/Y(t_j) ,$$

with

- ▶ $dN(t) = \sum_{i=1}^n dN_i(t)$ = no. observed events (or transitions) at time t
- ▶ $Y(t) = \sum_{i=1}^n Y_i(t)$ = no. at risk at time t
- ▶ Nelson-Aalen estimator

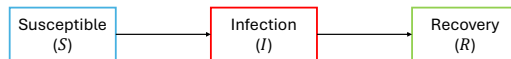
$$\hat{\Lambda}(t) = \sum_{0 < t_j \leq t} \Delta\hat{\Lambda}(t_j)$$

Estimation of intensities

Parametric estimates

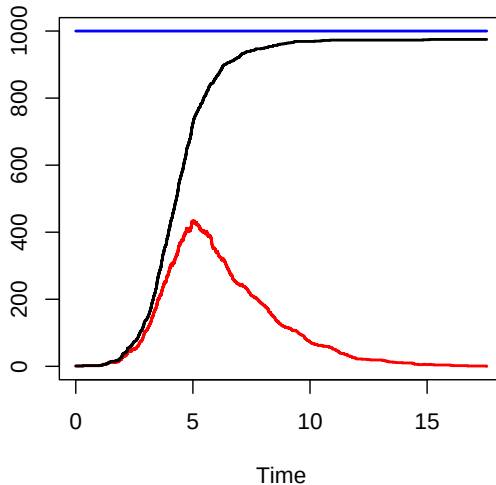
- ▶ Usually achieved by **maximum likelihood**
- ▶ Especially useful in case of **interval censoring**
- ▶ Constant rates
 - ▶ With right-censored (and left-truncated) data:
 - ▶ Estimate = Occurrence / Exposure (number of events / total follow-up) = $\int_0^\infty dN(t) / \int_0^\infty Y(t)dt$
 - ▶ SE of log(estimate) given by $1 / \sqrt{\text{no. of events}}$
 - ▶ With interval-censored data:
 - ▶ Likelihood based, feasible to express likelihood in terms of transition probabilities (coming up)
- ▶ Piecewise constant rates
- ▶ Poisson regression very useful in this context (together with splines for flexible parametric models)

Estimating the intensity of $S \rightarrow I$

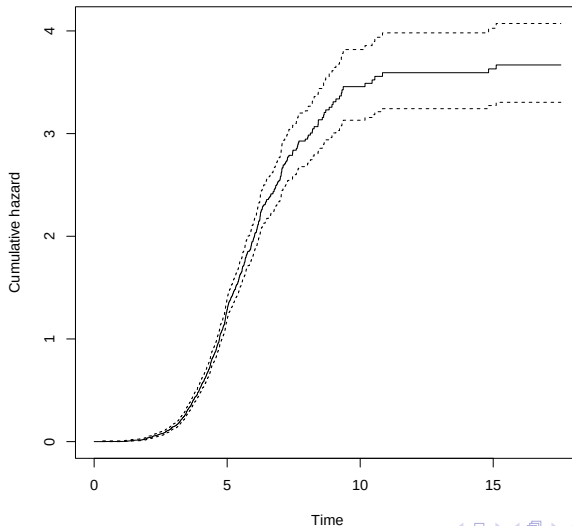


- ▶ Being at risk is the same as being susceptible
 - ▶ $Y_i(t) = S_i(t); Y(t) = S(t)$

Example SIR epidemic ($\beta = 2, \gamma = 0.5$)



Cumulative hazards



Bringing in mechanistic / biologic knowledge

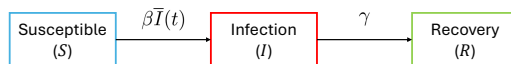
- ▶ So far estimated the (cumulative) intensity of infection (“force of infection”) non-parametrically, but we could bring in biological knowledge, for instance from the SIR model
- ▶ System of differential equations (overline means proportions)

$$\begin{aligned}\frac{d\bar{S}}{dt} &= -\beta\bar{S}(t)\bar{I}(t), \\ \frac{d\bar{I}}{dt} &= \beta\bar{S}(t)\bar{I}(t) - \gamma\bar{I}(t), \\ \frac{d\bar{R}}{dt} &= \gamma\bar{I}(t).\end{aligned}\tag{1}$$

- ▶ For the S → I transition, this means that the intensity can be written as

$$\lambda_i(t) = Y_i(t)\beta\bar{I}(t) = S_i(t)\beta\bar{I}(t)$$

SIR model

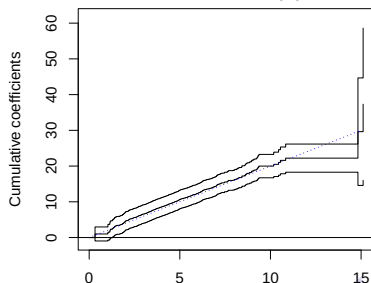


Cumulative $\beta(t)$

- Cumulative $\beta(t)$, say $B(t) = \int_0^t \beta(s)ds$, estimated as

$$\hat{B}(t) = \sum_{0 < t_j \leq t} \frac{dN(t_j)}{Y(t_j)\bar{I}(t_j)} = \sum_{0 < t_j \leq t} \frac{dN(t_j)}{S(t_j)\bar{I}(t_j)}$$

- Wolkewitz et al. (2002) used this idea for epidemic modeling with hospital outbreak data, and used martingale theory to prove consistency of $\hat{B}(t)$



Estimating time-constant β

- ▶ If we can assume that $\beta(t) \equiv \beta$, then constant β can be estimated using

$$\hat{\beta} = \frac{\sum_{0 < t_j \leq t} dN(t_j)}{\sum_{0 < t_j \leq t} S(t_j) \bar{I}(t_j)} = \frac{N(\infty)}{\int_0^\infty S(t) \bar{I}(t) dt}$$

- ▶ Similar to Occurrence / Exposure formula for constant hazards, but exposure is changed from $\int_0^\infty Y(t) dt$ to $\int_0^\infty Y(t) \bar{I}(t) dt$, a **weighted** exposure
- ▶ Obtained from maximizing the likelihood, see Becker & Britton (1999)

What about the I -> R transition?

- ▶ For the I -> R transition, according to SIR model, we get

$$\lambda_i(t) = I_i(t)\gamma$$

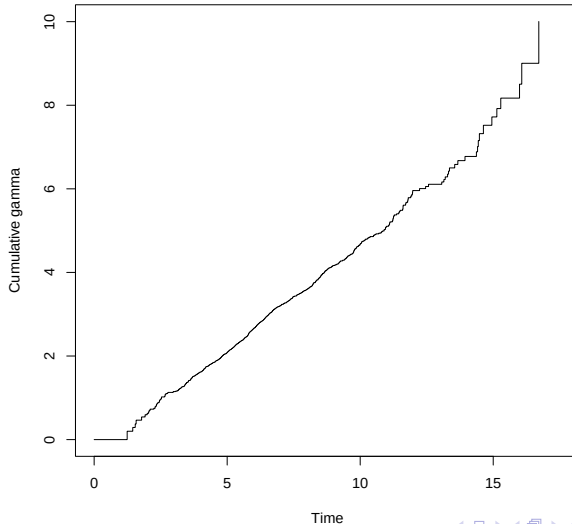
as the intensity of another counting process $R_i(t)$

- ▶ Constant recovery rate: estimate

$$\hat{\gamma} = \int_0^\infty dR(t) / \int_0^\infty I(t)dt$$

- ▶ If not constant, see discussion about time origin

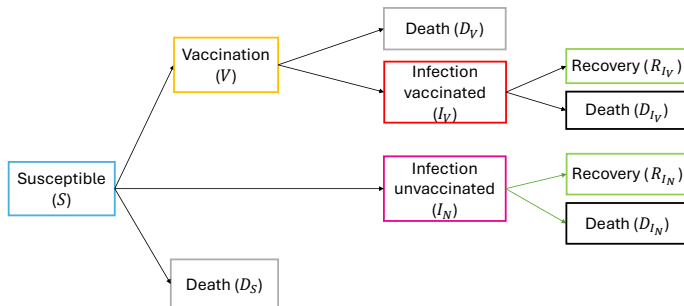
What about the I -> R transition?



Incomplete observation

- ▶ All of the formulas above assume complete observation (apart from right censoring and delayed entry)
- ▶ Not realistic
- ▶ Probably the biggest challenge in applying survival analysis to infectious disease modeling
- ▶ Tools from survival analysis combined with coarsening the observation scheme are possible
- ▶ My personal view: under-explored (TODO for survival analysis field)

What about more general multi-state models?



What about more general multi-state models?

- ▶ Broadly speaking, two approaches in modeling transition intensities in multi-state models
- ▶ Counting process approach (leads to step functions)
 - ▶ Implemented in {mstate} (de Wreede et al. 2011)
- ▶ Parametric approach (leads to smooth functions)
 - ▶ (Piecewise) constant hazards, {msm} package (Jackson, 2011)
 - ▶ Flexible parametric models
 - ▶ Often applied in interval censored multi-state models
- ▶ Need for hybrid approaches

Models incorporating covariates

Cox model

- ▶ Hazard given by

$$\lambda_i(t) = \lambda(t | Z_i(t)) = \lambda_0(t) \exp(\beta^\top Z_i(t))$$

- ▶ Well known and widely implemented in standard software
- ▶ Not discussed here in detail

Models incorporating covariates

Additive hazards model

- ▶ Hazard given by

$$\lambda_i(t) = \lambda(t | Z_i(t)) = \beta_0(t) + \beta^\top Z_i(t)$$

- ▶ Very useful in two settings
 - ▶ Relative survival (total rate is (known) background population rate + excess rate)

$$\lambda(t) = \lambda_p(t) + \lambda_e(t)$$

- ▶ Total force of infection driven by several sources of infection, naturally a sum

$$\lambda(t) = \beta S(t) \bar{I}(t) = S(t) \sum_k \beta_k \bar{I}_k(t)$$

Intensities versus probabilities

- ▶ Intensities express dynamically (incorporating the past) how events occur over time
- ▶ They usually are the primary target for inference and modeling
- ▶ But ultimately interest often is not only in the intensities / rates, but also in what these rates imply for a population, in terms of the proportion or numbers of infected individuals, hospitalized, vaccinated etc.
- ▶ So we want to go from intensities to probabilities

Standard survival analysis

- ▶ Relation between intensity and survival probability given by the **product integral**

$$S(t) = \prod_{0 < s \leq t} (1 - d\Lambda(s))$$

- ▶ This is the basis for the **Kaplan-Meier** estimate of the survival function

$$\hat{S}(t) = \prod_{0 < t_i \leq t} \left(1 - \frac{d_i}{Y_i}\right)$$

- ▶ Since $e^{-x} \approx 1 - x$ for small x , we also have

$$S(t) = \prod_{0 < s \leq t} \exp(-d\Lambda(s)) = \exp\left(-\int_0^t d\Lambda(s)\right) = \exp(-\Lambda(t))$$

- ▶ Traditionally the latter is used to obtain survival probabilities from the Cox model

More general settings

- ▶ In multi-state models we are interested in
 - ▶ State occupation probabilities

$$P(X(t) = \ell) := \pi_\ell(t), \quad \text{or} \quad P(X(t) \in \mathcal{L})$$

- ▶ Transition probabilities, conditioning on being in a particular state at present s

$$P(X(t) = \ell \mid X(s) = k) := P_{kl}(s, t)$$

- ▶ Complication: no longer one-to-one correspondence between rate and probability
 - ▶ Many transitions need to be taken into account at the same time
- ▶ One important assumption allows a mathematically very elegant theory: the **Markov assumption**

Markov property; Chapman-Kolmogorov

- ▶ The multi-state process is a **Markov process** if it fulfills the **Markov property**:

$$P(X(t) = \ell \mid X(s) = k, \mathcal{H}_{s-}) = P(X(t) = \ell \mid X(s) = k)$$

Future behavior of the process depends on the history only through the present

- ▶ Gather all transition probabilities in a matrix

$$\mathbf{P}(s, t) = \{P_{k\ell}(s, t)\}$$

- ▶ The transition probability matrices satisfy the **Chapman-Kolmogorov** equations:

$$\mathbf{P}(s, t) = \mathbf{P}(s, u)\mathbf{P}(u, t), \quad s < u < t$$

Kolmogorov equations

- ▶ Gather transition intensities $\lambda_{k\ell}(t)$, $\lambda_{kk}(t) = -\sum_{\ell \neq k} \lambda_{k\ell}(t)$, into $S \times S$ matrix $\lambda(t) = \{\lambda_{k\ell}(t)\}$
- ▶ Similarly for the cumulative transition hazards $\Lambda_{k\ell}(t)$ in $\Lambda(t)$
- ▶ $\mathbf{P}(s, t)$ is the unique solution of the **Kolmogorov forward differential equations**

$$\frac{\partial}{\partial t} \mathbf{P}(s, t) = \mathbf{P}(s, t) \lambda(t)$$

with initial condition $\mathbf{P}(s, s) = \mathbf{I}$

- ▶ By a general result for product-integrals (Volterra's equation), this solution takes the form

$$\mathbf{P}(s, t) = \prod_{(s, t]} (\mathbf{I} + d\Lambda(u))$$

State occupation probabilities

- ▶ For Markov models, general theory allows to express transition probabilities in terms of transition intensities
- ▶ State occupation probabilities can be obtained from those
- ▶ Define the vector

$$\pi(t) = (\pi_1(t), \dots, \pi_K(t))$$

- ▶ Then

$$\pi(t) = \pi(0)\mathbf{P}(0, t)$$

Estimating probabilities with constant hazards

- ▶ With constant hazards, define matrix λ

- ▶ Then

$$\mathbf{P}(s, t) = \text{expm}(\lambda(t - s)) = \mathbf{P}(0, t - s)$$

- ▶ Here $\text{expm}(A)$ is matrix exponential of A , given by

$$\text{expm}(A) = I + A + A^2/2 + \dots$$

- ▶ After having estimated each of the constant transition intensities, estimate of $\mathbf{P}(s, t)$ given by

$$\hat{\mathbf{P}}(s, t) = \text{expm}(\hat{\lambda}(t - s))$$

- ▶ Piecewise constant hazards: use $\hat{\mathbf{P}}(s, t) = \hat{\mathbf{P}}(s, u)\hat{\mathbf{P}}(u, t)$
- ▶ State occupation probabilities obtained through

$$\hat{\pi}(t) = \hat{\pi}(0)\mathbf{P}(0, t)$$

Estimating probabilities

Counting process approach

- ▶ With Nelson-Aalen estimates of the transition intensities, a reasonable estimator of $\mathbf{P}(s, t)$ is given by

$$\hat{\mathbf{P}}(s, t) = \prod_{s < u \leq t} (\mathbf{I} + \Delta \hat{\boldsymbol{\Lambda}}(u))$$

- ▶ where $\hat{\boldsymbol{\Lambda}}(u) = \{\hat{\Lambda}_{k\ell}(u)\}$ is a matrix with all estimated transition hazards and on the diagonal
$$\hat{\Lambda}_{kk}(u) = -\sum_{\ell \neq k} \hat{\Lambda}_{k\ell}(u)$$
- ▶ This is called the Aalen-Johansen estimator (1978)
- ▶ Generalizes Kaplan-Meier and non-parametric cumulative incidence estimate in competing risks

Scenario analyses

- ▶ Based on estimates / guesstimates / external “knowledge” of transition intensities (inputs)
- ▶ Some of these inputs are altered to express anticipated effects of interventions for instance
- ▶ Intervention could change $\hat{\Lambda}_{S \rightarrow I}(t)$ to $0.4\hat{\Lambda}_{S \rightarrow I}(t)$ (Aalen-Johansen based)
- ▶ Or it could change $\hat{\beta}I(t)$ to $0.4\hat{\beta}I(t)$ (ODE based)
- ▶ Hybrid (Aalen-Johansen and ODE based) versions useful
=> connecting survival analysis with infectious disease modeling

Discussion

- ▶ Not discussed: relative survival
- ▶ Not discussed: heterogeneity in susceptibility (= frailty in survival analysis)
- ▶ Not discussed: estimating distribution of serial interval, generation interval, incubation time

References



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